1 Scope

Does a standard or framework fix the scenarios of multi-party learning and state, for each, what a protocol hides and what its released output still leaks? The answer draws on the formalizing line's sweep of about 30 taxonomies, its ledger of 351 multi-party papers, and 9 searches of 2026-11-11 on standards and on frameworks that join the two.

2 Result

No standard or framework does it, and the place it would fill lies between three bodies of work, each holding one part. Taxonomies give the vocabulary of stage, goal, knowledge and capability but prove nothing, and none has an axis for which parties the adversary controls. Security games make exact what the adversary can tell, but Salem et al. [16] state that their 18 games have one challenger training one model and do not express concurrent protocols. Simulation proofs show that a protocol reveals nothing beyond the output of an ideal functionality [10, 9], and they take what that output leaks as given. The two layers meet only inside single constructions, in 12 of the 48 ledger papers proven against an ideal functionality, each for its own protocol. Standards for federated learning, MPC, FHE and DP fix vocabulary and the unit of privacy, and none fixes attack scenarios.

Does not show: Only Salem et al., Yeom et al. and HE-OFT were read in full. The other entries rest on sweep notes, abstracts or search snippets, and a limit marked as found by us may be wrong in detail. The ledger counts come from language-model coders of 2026-10-16 and were not re-read, and the full text of IEEE 3652.1 was not seen.

3 How it was produced

Sources	the taxonomy sweep of 2026-08-05 and the ledger and notes of the formalizing line, all in `FORMAL_SECURITY`, and web searches
Range	works from 2010 to 2026
Queries	9 searches, logged by slug in the Searches section of `lib/literature/positioning.md`
Screened	the 351 ledger papers counted by formality, and every row of `lib/literature/screening.csv`
Artifacts	the CSVs under this report's stem in `lib/reports/src/data/`
Reproduce	the tables script in `lib/reports/src/figures/`, run from the HE-ML root with FORMAL_SECURITY beside it

4 What this changes

The framework Halil asked for is unoccupied, and the paper can therefore claim the framework itself if it reuses the standard objects of the three bodies. Positioning against Villa et al. [18], which the sweep of 2026-08-05 named the closest work, is no longer needed, because its object is an agentic LLM system. The closest prior work becomes Salem et al. for the output layer, Li et al. [8] for games with several parties, and secure aggregation for the protocol layer. The 12 joined constructions are instances the framework must reproduce as special cases.

5 Next

Read Salem et al. and Li et al. side by side, then the simulation definition of Flamingo, and agree how a scenario is written before any definition is fixed.

Table 1. The bodies of prior work, what each gives a framework of scenarios, and where each stops. Found by us means our reading, not a statement of the authors.

Body	Works	What it gives us	Where it stops
Taxonomies	Barreno et al. [1], Papernot et al. [13], NIST AI 100-2 [17]	The terms for the stage, the goal, the adversary's knowledge and its capability.	They prove nothing. None has an axis for which parties the adversary controls (found by us). Only NIST puts privacy and integrity on one page.
Federated SoKs	Carletti et al. [2], Mansouri et al. [10], Rizzini et al. [15]	Threat models by role inside one setting, such as eight server models for gradient inversion, one a server with Sybil clients.	Each covers one setting and one family of attacks (found by us).
Security games	Yeom et al. [20], Salem et al. [16], Wu et al. [19], Li et al. [8]	What it means that the adversary can tell, for a released model or for gradients, with reductions between goals.	Salem et al.: one challenger trains one model, and games do not express concurrent protocols (stated, Sec. VI-D). Wu et al.: integrity only, with no advantage. Li et al.: gradients only, the final model is not attacked.
Protocol simulation	Mansouri et al. [10], Ma et al. [9], HE-OFT [7], Morona-Mínguez et al. [11]	The protocol reveals nothing beyond the output of an ideal functionality, against a named coalition.	What that output leaks is taken as given, for instance the sum in secure aggregation. A CKKS instantiation is approximate, so it realizes the functionality only up to an error.
Protocol with a mechanism on the output	Pentyala et al. [14], and the joined ledger papers	MPC or HE for the protocol and DP on the released value, in one construction.	Each proof covers its own protocol and functionality. No search found a definitional framework.
Games in another domain	Villa et al. [18]	One game for confidentiality, integrity and availability.	Its object is an agentic LLM system, not parties training together.
Standards	IEEE 3652.1 [4], ISO/IEC 4922-1 [5], ISO/IEC 28033-1 [6], NIST SP 800-226 [12]	Vocabulary for federated learning, MPC, FHE and DP, and the unit of privacy.	None fixes attack scenarios. ISO/IEC 28033 is a draft, and the full text of IEEE 3652.1 was not seen.
Method	Desfontaines and Pejó [3]	Criteria for the dimensions of a taxonomy: inner exclusivity and mutual compatibility.	Applied to variants of DP, not to attack scenarios.

Table 2. The gap, item by item, with the evidence and whose finding it is.

Gap	What is missing	Evidence	Whose finding
no-scenario-coordinates	No work writes a scenario as the parties, the ideal functionality and the coalition, with a security statement per scenario.	The sweep of roughly thirty taxonomies of 2026-08-05 and the searches of 2026-11-11, all nil.	found by us
games-one-model	The game frameworks have one challenger training one model.	Salem et al. [16], Sec. VI-D.	stated
simulation-stops-at-output	Simulation proofs say nothing about what the functionality's output leaks.	The ideal leakage of secure aggregation is the sum [10].	stated, as the definition
joins-per-construction	Protocol security and output leakage are joined only inside single constructions.	The joined papers of the ledger, and Pentyala et al.	found by us
privacy-integrity-apart	Privacy and integrity are formalized apart.	Salem et al. exclude integrity, Wu et al. [19] exclude privacy.	found by the 2026-08-05 sweep
no-standard-scenarios	No standard fixes attack scenarios.	The four standards above.	found by us

Table 3. The formality of the multi-party papers of the ledger, as the sweep coders recorded it on 2026-10-16. One paper counts once, by its first coded threat model.

Formality	Papers	What such a paper proves
theorem with assumptions	161	a bound on its own method under its own assumptions
prose	118	an argument with no theorem
ideal functionality	48	that its protocol reveals nothing beyond a stated ideal functionality
game	16	an advantage bound in a game it defines
taxonomy	8	a classification

Table 4. The ledger papers proven against an ideal functionality that also put a DP mechanism on the released value, as coded on 2026-10-16. A key longer than fourteen letters is cut.

Key	Venue	Setting
ALZK25	PoPETs 2025	MPC training (two parties, FHE with label DP)
BKP25	ICML 2025	secure aggregation (distributed DP for horizontal FL)
CPARLS26	IEEE S&P 2026	MPC training (two-party HE, vertical label partition)
FFRWJPD25	CCS 2025	MPC training, secure aggregation
FGSW26	USENIX Security 2026	MPC training (DP synthetic tabular data, vertical)
HMSS25	ASIACRYPT 2025	multiparty HE (two-party sender-receiver protocol on single-key CKKS, setting set by the batch rule)
JRWNC26	PoPETs 2026	vertical FL (MPC training with DP)
LLLXX24	NDSS 2024	MPC training (two-party secret sharing with HE and OT, n data owners reduced to sequential two-party interaction)
TH26	PoPETs 2026	secure aggregation (cross-device private FL, DP-FTRL with TEE)
TWMJFBBCCC...	CCS 2024	secure aggregation (two-server, Prio-based, private federated learning and analytics)
XZH24	CCS 2024	MPC training (three-server shuffle model of DP, horizontal FL)
XZH25	CCS 2025	MPC training (three-server secure aggregation, horizontal FL)

References

[1] Marco Barreno, Blaine Nelson, Anthony D. Joseph, and J. D. Tygar. The security of machine learning. *Machine Learning*, 2010.

[2] Vincenzo Carletti, Pasquale Foggia, Carlo Mazzocca, Giuseppe Parrella, and Mario Vento. SoK: Gradient inversion attacks in federated learning. In *USENIX Security Symposium*, 2025.

[3] Damien Desfontaines and Balázs Pejó. SoK: Differential privacies. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2020.

[4] IEEE. IEEE guide for architectural framework and application of federated machine learning. IEEE Std 3652.1-2020, 2020.

[5] ISO/IEC. Information security, secure multiparty computation, part 1: General. ISO/IEC 4922-1:2023, 2023.

[6] ISO/IEC. Fully homomorphic encryption, part 1: General. ISO/IEC DIS 28033-1, draft, 2026.

[7] Halil İbrahim Kanpak, Sinem Sav, and Alptekin Küpçü. HE-OFT: Privacy-preserving one-shot federated fine-tuning under homomorphic encryption. arXiv:2610.08255, 2026.

[8] Zhuohang Li, Andrew Lowy, Jing Liu, Toshiaki Koike-Akino, Kieran Parsons, Bradley Malin, and Ye Wang. Analyzing inference privacy risks through gradients in machine learning. In *ACM Conference on Computer and Communications Security (CCS)*, 2024.

[9] Yiping Ma, Jess Woods, Sebastian Angel, Antigoni Polychroniadou, and Tal Rabin. Flamingo: Multi-round single-server secure aggregation with applications to private federated learning. In *IEEE Symposium on Security and Privacy (S&P)*, 2023.

[10] Mohamad Mansouri, Melek Önen, Wafa Ben Jaballah, and Mauro Conti. SoK: Secure aggregation based on cryptographic schemes for federated learning. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2023.

[11] Miguel Morona-Mínguez, Fernando Pérez-González, and Alberto Pedrouzo-Ulloa. Pushing forward multi-secret-key homomorphic encryption for private average aggregation. arXiv:2609.01945, 2026.

[12] Joseph P. Near and David Darais. Guidelines for evaluating differential privacy guarantees. Technical Report SP 800-226, NIST, 2025.

[13] Nicolas Papernot, Patrick McDaniel, Arunesh Sinha, and Michael P. Wellman. SoK: Security and privacy in machine learning. In *IEEE European Symposium on Security and Privacy (EuroS&P)*, 2018.

[14] Sikha Pentyala, Davis Railsback, Ricardo Maia, Rafael Dowsley, David Melanson, Anderson Nascimento, and Martine De Cock. Training differentially private models with secure multiparty computation. arXiv:2202.02625, 2022.

[15] Andrea Rizzini, Marco Esposito, Tommaso Gagliardoni, and Francesco Bruschi. SoK: Verifiable integrity claims for privacy-preserving federated learning. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2026.

[16] Ahmed Salem, Giovanni Cherubin, David Evans, Boris Köpf, Andrew Paverd, Anshuman Suri, Shruti Tople, and Santiago Zanella-Béguelin. SoK: Let the privacy games begin! A unified treatment of data inference privacy in machine learning. In *IEEE Symposium on Security and Privacy (S&P)*, 2023.

[17] Apostol Vassilev, Alina Oprea, Alie Fordyce, Hyrum Anderson, Xander Davies, and Maia Hamin. Adversarial machine learning: A taxonomy and terminology of attacks and mitigations. Technical Report AI 100-2e2025, doi:10.6028/NIST.AI.100-2e2025, NIST, 2025.

[18] Federico Villa, F. Betül Durak, Tadayoshi Kohno, Tapdig Maharramli, and Franziska Roesner. Extending the formalism and theoretical foundations of cryptography to AI. arXiv:2603.02590, 2026.

[19] Baoyuan Wu, Zihao Zhu, Li Liu, Qingshan Liu, Zhaofeng He, and Siwei Lyu. Attacks in adversarial machine learning: A systematic survey from the life-cycle perspective. arXiv:2302.09457, 2023.

[20] Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. Privacy risk in machine learning: Analyzing the connection to overfitting. In *IEEE Computer Security Foundations Symposium (CSF)*, 2018.